



Lecture 7

- Gauss's law examples

Gauss's law



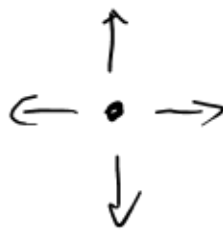
Differential form :
(Point-based)

electric flux density $\left[\frac{C}{m^2}\right]$

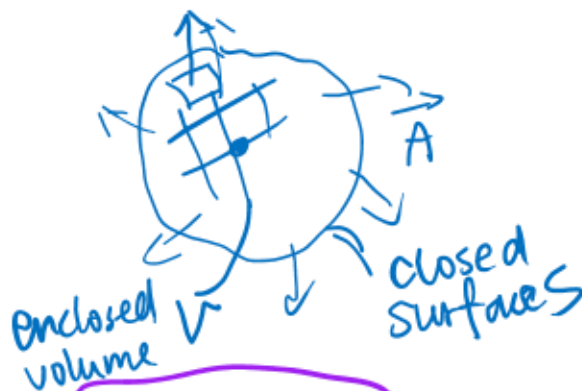
$$\nabla \cdot \vec{D} = \rho_v$$

volume charge density $\left[\frac{C}{m^3}\right]$

$$\vec{D} = \epsilon \vec{E}$$



Divergence Theorem



$$\oint_S \vec{A} \cdot d\vec{S} = \int_V \nabla \cdot \vec{A} dv$$

Integral form

flux of $\vec{A}$ thru the closed surface

$$\oint \vec{D} \cdot d\vec{S} = \int_V \nabla \cdot \vec{D} dv = \int_V \rho_v dv = Q_{\text{enclosed}}$$

Q_{enc}
total charge enclosed by the surface S.

Gauss's law

$$\oint_S \vec{D} \cdot d\vec{S} = Q_{\text{enc}} = \int_V \rho_v dv$$

Steps:

1. Check the symmetry of the charge – **critical**

Make sure the charge distribution has sufficient symmetry so that you know the **direction of D** and can find a surface on which its **magnitude is constant**.

2. Choose that Gaussian surface

3. Evaluate the left-hand side: $\oint_S \vec{D} \cdot d\vec{S}$

4. Evaluate the right-hand side: $Q_{\text{enc}} = \int_V \rho_v dv$

5. Find **D**, or **E**

$$\mathbf{E} = \frac{\mathbf{D}}{\epsilon}$$

Move the observation point using that rotation/translation, and ask whether the new point is physically equivalent to the original point because the charge distribution looks exactly the same from there.

Three standard Gauss-law symmetries

Charge distribution

Symmetry test

Point charge / spherical distribution

Rotate about the center → unchanged

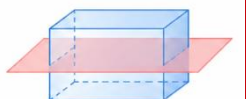
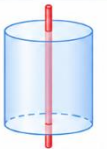
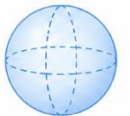
Infinite line charge

Rotate around line **and translate along line**
→ unchanged

Infinite plane charge

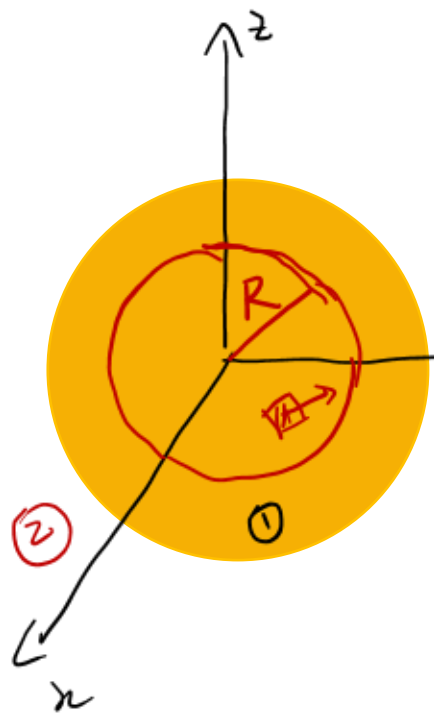
Translate anywhere parallel to plane → unchanged

Example Gaussian surface



Gauss's law

Example 1: A sphere of radius a is charged with uniform volume charge density ρ_{v0} . Calculate $\vec{D}$ and $\vec{E}$ everywhere in space. Assume ϵ_0 throughout.



$$\vec{D} = \hat{R} D_R(R)$$

$$\rho_v^{(R)} = \begin{cases} \rho_{v0} & , R \leq a \\ 0 & , R > a \end{cases}$$

$$\text{LHS: } \oint_S \vec{D} \cdot d\vec{S}$$

$$\hat{R} D_R$$

$$\hat{R} R^2 \sin\theta d\theta d\phi$$

$$\text{LHS} = D_R \underbrace{\int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} R^2 \sin\theta d\theta d\phi}_{4\pi R^2} = D_R \cdot 4\pi R^2$$

$$\text{Region 1: } R \leq a$$

$$\text{RHS: } Q_{\text{enc}}$$

$$= \int_V \rho_v dv$$

$$= \rho_v \cdot \int_V dv$$

$$= \rho_v \cdot \frac{4\pi R^3}{3}$$

$$D_R \cancel{4\pi R^2} = \rho_v \frac{4\pi R^3}{3}$$

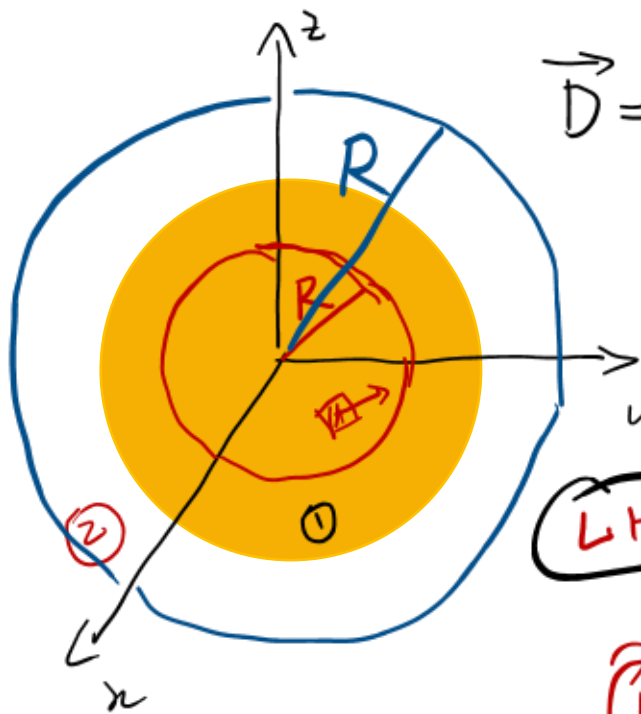
$$D_R = \rho_v \frac{R}{3}$$

$$E_R = \frac{D_R}{\epsilon_0} = \frac{\rho_v R}{3\epsilon_0}$$



Gauss's law

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$$\vec{D} = \hat{R} D_R(R)$$

$$\rho_v^{(R)} = \begin{cases} \rho_{v0} & , R \leq a \\ 0 & , R > a \end{cases}$$

LHS

$$\oint_S \vec{D} \cdot d\vec{S}$$

$$\hat{R} D_R$$

$$\hat{R} R^2 \sin\theta d\theta d\phi$$

$$\text{LHS} = D_R \underbrace{\int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} R^2 \sin\theta d\theta d\phi}_{4\pi R^2} = D_R \cdot 4\pi R^2$$

W

Region 2: $R > a$

RHS: Q_{enc}

$$= \int_V \rho_v dv \quad \begin{matrix} R=a \\ R=0 \end{matrix}$$

$$= \rho_v \cdot \int_V dv$$

$$= \rho_v \cdot \frac{4\pi a^3}{3}$$

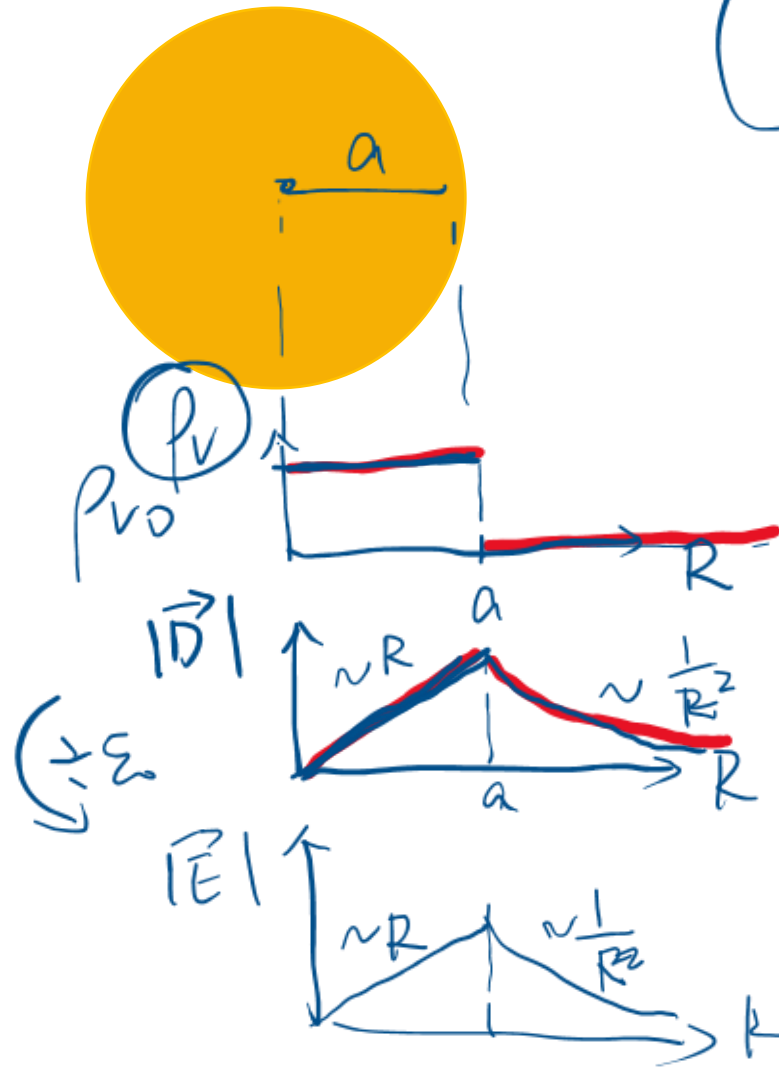
$$D_R \cdot 4\pi R^2 = \rho_v \frac{4\pi a^3}{3}$$

$$\left\{ \begin{array}{l} D_R = \frac{\rho_v a^3}{3R^2} \\ E_R = \frac{\rho_v a^3}{3\epsilon_0 R^2} \end{array} \right.$$

Example 1: A sphere of radius a is charged with uniform volume charge density ρ_{v0} . Calculate $\vec{D}$ and $\vec{E}$ everywhere in space. Assume ϵ_0 throughout.

$$\vec{D} = \begin{cases} \hat{R} \rho_{v0} \frac{R}{3} & R \leq a \\ \hat{R} \rho_{v0} \frac{a^3}{3R^2} & R > a \end{cases} \quad \Leftarrow$$

$$\vec{E} = \begin{cases} \hat{R} \rho_{v0} \frac{R}{3\epsilon_0} & R \leq a \\ \hat{R} \rho_{v0} \frac{a^3}{3\epsilon_0 R^2} & R > a \end{cases} \quad \left(\frac{1}{R^2} \right)$$



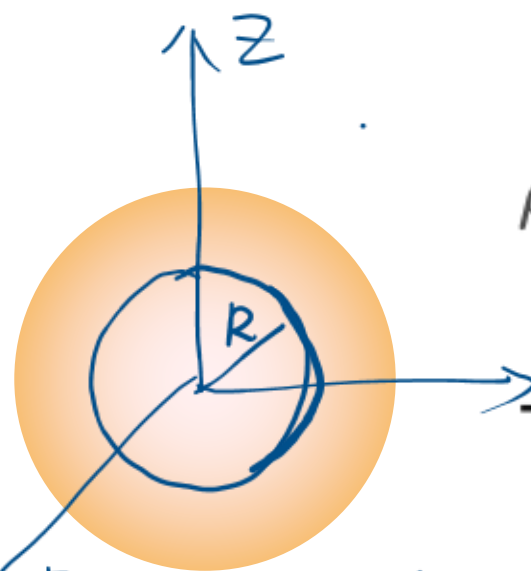
Example 2: A sphere of radius a , with

Gauss's law

$$\rho_v(R) = \begin{cases} \rho_{v0} \left(\frac{R}{a}\right)^2, & R \leq a \\ 0, & R > a \end{cases}$$

$$D_R = \cancel{4\pi R^2} = \rho_{v0} \cdot \frac{R^3 \cdot 4\pi}{5a^2}$$

$$D_R = \rho_{v0} \frac{R^3}{5a^2} \quad E_R = \rho_{v0} \frac{R^3}{5\epsilon a^2}$$



The sphere has permittivity ϵ and is surrounded by ϵ_0 .

Region 1 $R \leq a$

$$\oint_S \vec{D} \cdot d\vec{S} = \oint_S R D_R \cdot R^2 \sin\theta d\theta d\phi \hat{R}$$

$$= D_R 4\pi R^2$$

$$= \frac{\rho_{v0}}{a^2} \frac{R^5}{5} \cdot 2 \cdot 2\pi$$

$$Q_{enc} = \int_V \rho_v dv$$

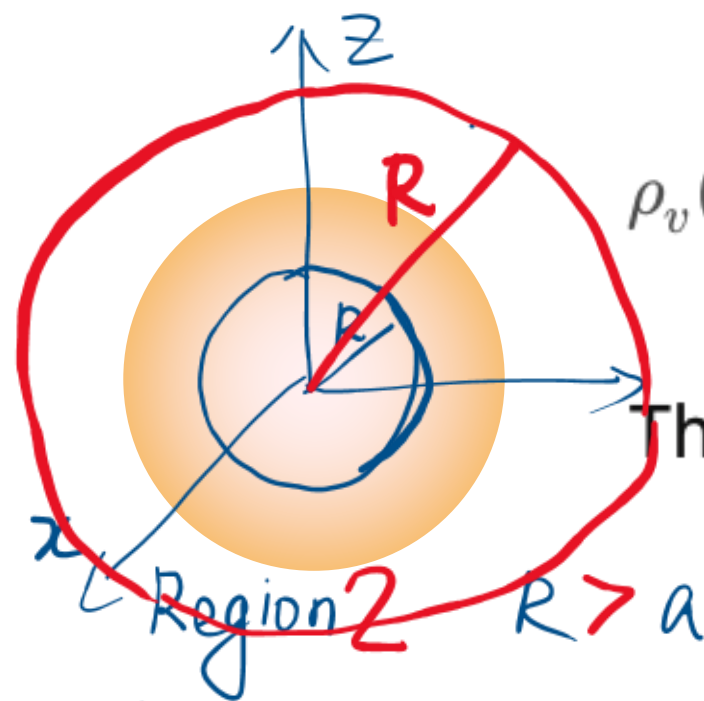
$$= \int_V \rho_{v0} \left(\frac{R}{a}\right)^2 \cdot R^2 \sin\theta dR d\theta d\phi$$

$$= \int_{R=0}^R \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \rho_{v0} \frac{R^4}{a^2} \sin\theta dR d\theta d\phi$$

$$= \frac{\rho_{v0}}{a^2} \int_{R=0}^R R^4 dR \int_{\theta=0}^{\pi} \sin\theta d\theta \int_{\phi=0}^{2\pi} d\phi$$

Example 2: A sphere of radius a , with

Gauss's law



$$\rho_v(R) = \begin{cases} \rho_{v0} \left(\frac{R}{a}\right)^2, & R \leq a \\ 0, & R > a \end{cases}$$

The sphere has permittivity ϵ and is surrounded by ϵ_0 .

$$D_R 4\pi R^2 = \frac{\rho_{v0} \cdot a^3 \cdot 4\pi}{5}$$

$$D_R = \frac{\rho_{v0} a^3}{5R^2} \quad E_R = \frac{D_R}{\epsilon_0} = \frac{\rho_{v0} a^3}{5\epsilon_0 R^2}$$

$$\oint_S \vec{D} \cdot d\vec{S} = \oint_S R D_R \cdot R^2 \sin\theta d\theta d\phi \hat{R}$$

$$= D_R 4\pi R^2$$

$$= \frac{\rho_{v0}}{a^2} \frac{a^5}{5} \cdot 2 \cdot 2\pi = \frac{\rho_{v0} a^3 \cdot 4\pi}{5}$$

$$= \frac{\rho_{v0}}{a^2} \int_{R=0}^a R^4 dR \int_{\theta=0}^{\pi} \sin\theta d\theta \int_{\phi=0}^{2\pi} d\phi$$

$$Q_{enc} = \int_V \rho_v dv$$

$$= \int_V \rho_{v0} \left(\frac{R}{a}\right)^2 \cdot R^2 \sin\theta dR d\theta d\phi$$

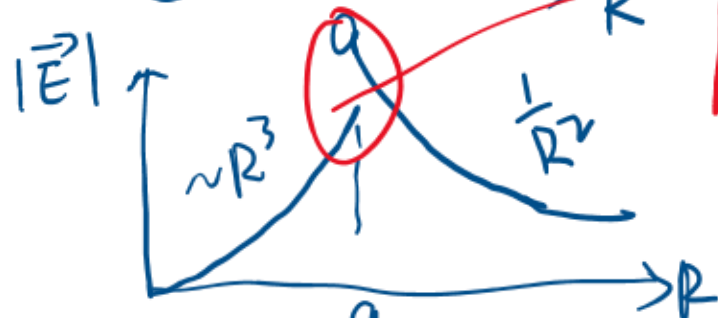
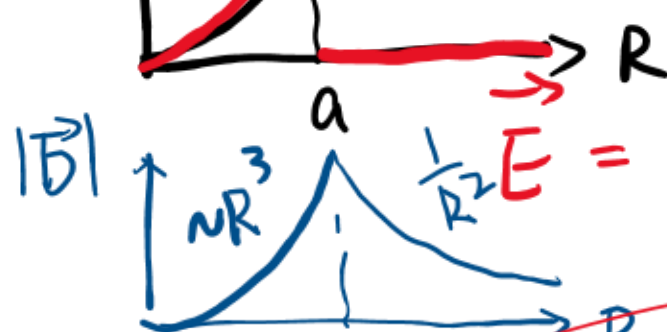
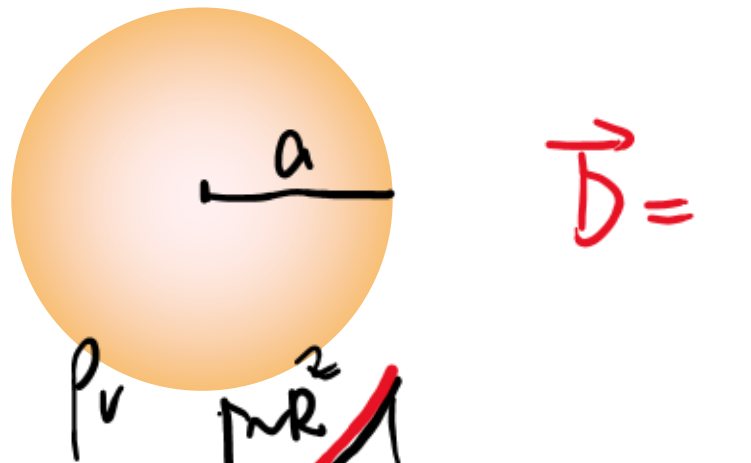
$$= \int_{R=0}^a \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \rho_{v0} \frac{R^4}{a^2} \sin\theta dR d\theta d\phi$$

Example 2: A sphere of radius a , with

Gauss's law

$$\rho_v(R) = \begin{cases} \rho_{v0} \left(\frac{R}{a}\right)^2, & R < a \\ 0, & R > a \end{cases}$$

The sphere has permittivity ϵ and is surrounded by ϵ_0 .



$$\vec{D} = \begin{cases} \hat{R} \frac{\rho_{v0} R^3}{5a^2} & R \leq a \\ \hat{R} \frac{\rho_{v0} a^3}{5R^2} & R > a \end{cases}$$

$$\vec{E} = \begin{cases} \hat{R} \frac{\rho_{v0} R^3}{5\epsilon a^2} & R \leq a \\ \hat{R} \frac{\rho_{v0} a^3}{5\epsilon_0 R^2} & R > a \end{cases}$$

discontinuity of $\vec{E}$ is because discontinuity of material permittivity

Gauss's law

Example 3: An infinitely long cylinder of radius a is charged with uniform volume charge density ρ_{v0} . Assume $\epsilon = \epsilon_0$ throughout. Find $\vec{E}$ everywhere in space.



$$\rho = \begin{cases} \rho_{v0} & , r \leq a \\ 0 & , r > a \end{cases}$$

$$\vec{D} = \hat{r} D_r(r)$$

Region 1: $r \leq a$

$$\oint_S \vec{D} \cdot d\vec{S} = \int_S \vec{D} \cdot d\vec{S} + \int_S \vec{D} \cdot d\vec{S} + \int_S \vec{D} \cdot d\vec{S}$$

$\hat{r} D_r$ $\hat{r} r d\phi dz$

$$D_r \oint_S r d\phi dz = D_r \cdot 2\pi r h$$

RHS

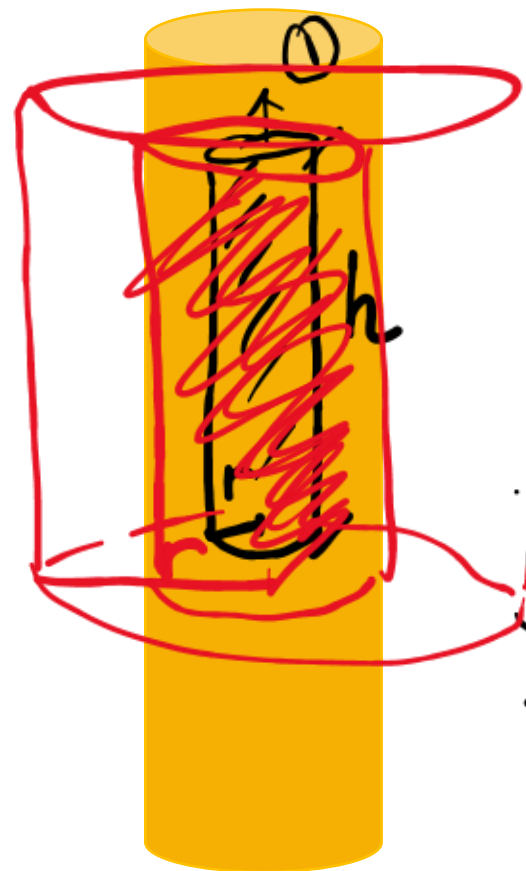
$$\int_V \rho_v dv = \rho_{v0} \pi r^2 h$$

$$D_r = \frac{r \rho_{v0}}{2}$$

$$E_r = \frac{r \rho_{v0}}{2\epsilon_0}$$

Gauss's law

Example 3: An infinitely long cylinder of radius a is charged with uniform volume charge density ρ_{v0} . Assume $\epsilon = \epsilon_0$ throughout. Find $\vec{E}$ everywhere in space.



$$\rho = \begin{cases} \rho_{v0} , & r \leq a \\ 0 , & r > a \end{cases}$$

$$\vec{D} = \hat{r} D_r(r)$$

Region 2 - $r > a$

$$\oint_S \vec{D} \cdot d\vec{S} = \int_S \vec{D} \cdot d\vec{S} + \int_S \vec{D} \cdot d\vec{S} + \int_S \vec{D} \cdot d\vec{S}$$

$\hat{r} D_r$ $\hat{r} r d\phi dz$

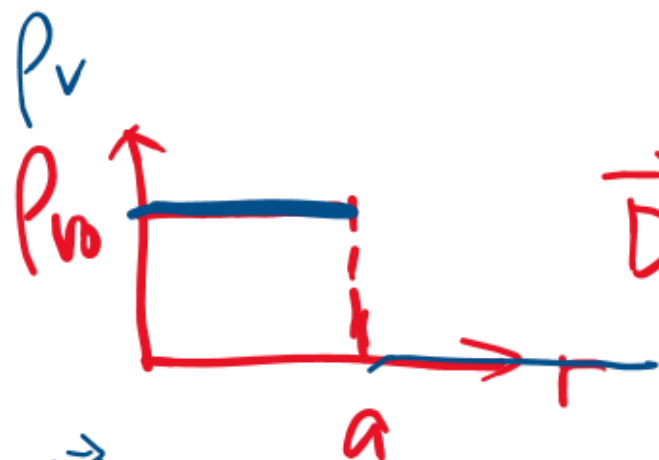
$$D_r \oint_S r d\phi dz = D_r \cdot 2\pi r h$$

RHS

$$\int_V \rho_v dv = \rho_{v0} \pi a^2 h$$

$$D_r = \frac{\rho_{v0} a^2}{2r}$$

$$E_r = \frac{\rho_{v0} \cdot a^2}{2\epsilon_0 r}$$

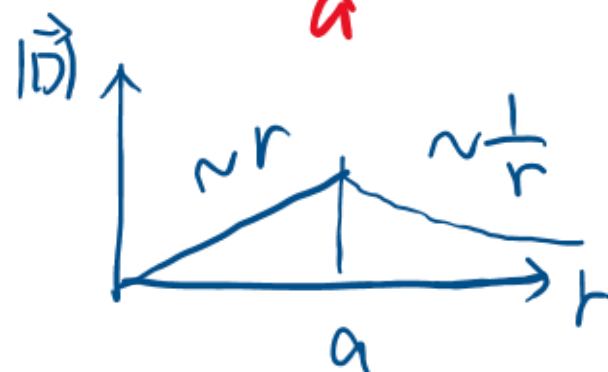


$\vec{D} =$

$$\begin{cases} \hat{r} \rho_{v0} \frac{r}{2} & r \leq a \\ \hat{r} \rho_{v0} \frac{a^2}{2r} & r > a \end{cases}$$

$r \leq a$

$r > a$



$\vec{E} =$

$$\begin{cases} \hat{r} \rho_{v0} \frac{r}{2\epsilon_0} & r \leq a \\ \hat{r} \rho_{v0} \frac{a^2}{2\epsilon_0 r} & r > a \end{cases}$$

$r \leq a$

$r > a$

$\frac{1}{r}$

